

Prediction of Waiting Time in Queues: An Ensemble Learning Approach

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Abstract—Queuing up for a service is sometimes an inevitable experience. The inefficiencies brought on by extended waiting times can be considerably decreased by precise waiting time prediction. Accurate prediction can substantially improve consumer satisfaction by reducing uncertainty. It is possible to introduce a robust approach to the prediction of waiting times based on previous queuing data and artificial intelligence (AI) algorithms. This paper contributes to the field by offering a robust approach to waiting time prediction and suggests potential directions for further research. The investigation leverages ensemble tree-based methods, supplemented by various data pre-processing techniques for regression analysis to forecast precise waiting times. The following regression models have been used to assess the performance: Random Forest (RF), Extra Trees (ET), Gradient Boosting (GBR), Histogram-Based Gradient Boosting (HGBR), and Voting (VR). Among these, the Extra Trees Regressor demonstrates superior performance. Dimensionality reduction via Principal Component Analysis (PCA) proved less effective than using the original feature sets. Furthermore, the challenge of data imbalance in classification tasks has also been addressed here using the Synthetic Minority Oversampling Technique (SMOTE). This process impressively enhances classification accuracy, especially for minority classes.

I. INTRODUCTION

Managing a queue in a First-come First-served (FCFS) manner is the most straightforward approach. This strategy of processing requests in the exact order of their arrival is particularly effective in scenarios where all clients are treated as equals ensuring a fair and impartial service [1]. In our study, we have utilized a dataset from a banking service where the queuing process was implemented using the same FCFS approach [2]. In any service, the waiting time in a queue plays a big role in client satisfaction. In real-life scenarios, waiting in a queue is often unavoidable. However, probable waiting time forecast before standing up in a queue can greatly lessen the discontentment that comes with waiting for a lengthy period. So far, numerous approaches have been investigated in the field of waiting time prediction to enhance the precision and effectiveness of service-oriented systems. A prominent method

was put in 1999 where customer waiting time was statistically measured based on several parameters like queue length, server count, and average service rates [3]. In the meantime, improvements in accuracy have been obtained with an improved version of the queue theory-based predictor that integrates the renege rate into the prediction model [4]. Later on, researchers presented a historical-based predictor that relies on previous clients' waiting times [5], [6]. Waiting time prediction has also been approached using several advanced techniques like quantile regression and regression splines. Regression splines were used in multi-skill call center scenarios with waiting periods and server counts as inputs [7]. Another study investigated quantile regression, concentrating on particular response time percentiles to forecast wait times, especially in hospital emergencies [8]. Such a prediction with Artificial Neural Networks (ANN) has also been proven to be a powerful tool due to its complex modeling and parameter optimization [7]. However, a key limitation of ANNs, especially deep neural networks, is their difficulty with generalization. Though the performance of ANN is quite satisfactory, they are constrained by their black-box nature and lack of hierarchical perception. Training ANNs, particularly deep learning models, is often time-consuming and requires significant computational resources. The process is also energy-intensive, raising sustainability concerns [9], [10]. One noteworthy study used Support Vector Regression (SVR) in the analysis of a queue dataset consisting of real-life data from multiple banks [11]. The study predicted waiting times for individuals and compared the performance of SVR with other machine learning techniques such as K-Nearest Neighbors (KNN) and K-Means Clustering. Though the predictions with SVR is quite impressive compared to different methods, we need to be aware that, SVR can distort the ϵ -tube and lead to overfitting which can outcome poor generalization on unseen data [12]. Besides, SVR's computational complexity is an issue with large datasets, as training involves solving a quadratic programming problem [13]. Here, we investigate waiting time prediction on an existing set of data in Bank Teller Lines. Research has already been found on existing datasets to utilize several models like Queuing Theory (QT), Deep Learning (DL), Gradient Boosting Machine (GBM), and Random Forest (RF) [2]. Results revealed that the GBM model has the highest accuracy (97%) and F1-measure (75%). However, their study focused solely on time overflow prediction, which determines whether a client waited for more than 30 minutes. The current study takes a fresh look at the analysis of possible waiting periods for a client by including numerical forecasting in the prediction process. Such a method utilizes regression analysis which is typically used for predicting

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the exact value of a target variable. We opt for ensemble tree algorithms because conventional regression methods, such as linear regression, have limitations in capturing nonlinear relationships in the data [14]. Since we aim to predict waiting time, we feel it is more appropriate to forecast the numerical value of time rather than a discrete output with the help of classification algorithms. To identify the optimal feature sets, we employ feature selection techniques with forward selection and backward elimination. Concurrently, using the same dataset, we explore a classification approach by categorizing the entire waiting time range into several groups. To rectify the underlying imbalance in class, the study utilizes the Synthetic Minority Over Sampling Technique (SMOTE) [15]. Finally, we evaluate the performance of several classifier algorithms on this balanced dataset.

II. WAITING TIME PREDICTION

This study employs a machine learning approach that is an effective tool for analyzing and predicting purposes using complex datasets [15]. As shown in Figure 1, the process begins with the collection of the dataset. Following this, several steps of data preprocessing and feature selection are performed. The refined dataset is then fed into some selected models which are trained using the dataset. Subsequently, the performance has been evaluated using various metrics.

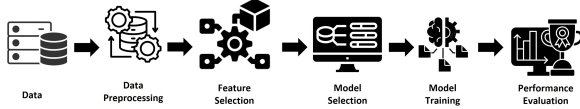


Fig. 1: Flowchart of waiting time prediction process

A. Dataset

The dataset for this study has been taken from archived teller logs of a Brazilian bank [2]. The logs cover one month, from August 1st, 2016, to August 31st, 2016, and contain a total of 12,708 data points. Each data point consists of 14 features, along with a target variable named ‘waiting_time’. Detailed explanations for all the columns are not available, which makes it challenging to properly understand the statistics. There is a feature column called ‘service_type’ blending four categories: A, B, C, and D. However, the definition of A, B, C, and D are not available in the dataset. Despite the lack of complete information, this dataset aids in our analysis of customer service trends.

B. Data Preprocessing

Raw data often contain irregularities like missing values, noise, inconsistencies, and redundancies, which can affect the model’s performance [16]. Besides, preprocessing of categorical data is essential, as most machine learning models only accept numerical variables. Here, the data pre-processing has been performed with the following operations.

- **Null cancellation:** The ‘client_id’ column has been discarded due to the presence of null values. Presence of

Feature Name	Description
branch_id	Unique IDs of respective branches
date_of_service	Date
queue_id	Unique ID of each queue
sequence	Serial number of the client in the corresponding queue
status	Whether service is finished or reneged
entity	Whether a client is a person or a company
priority	TRUE if the client has any priorities; otherwise, FALSE
arrival_time	Joining time in the queue
start_time	Service start time
end_time	Service end time
client_id	Unique ID number of the clients
cashier	Unique ID number of the bank personnel who served
service_type	Services that clients received, categorized by A, B, C, and D
waiting_time	The amount of time clients required to wait before receiving a service

TABLE I: Dataset feature description

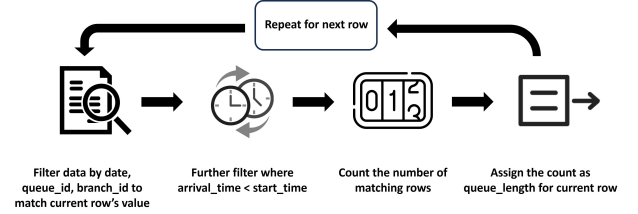


Fig. 2: Queue length calculation process

null values can introduce bias, complicate pre-processing, and hence significantly affect model interpretation and performance [17].

- **Time conversion:** All time-related features have been converted to minutes because it provides a good balance of detail and simplicity compared to seconds or hours when measuring time-related features [18].
- **Label encoding:** Label encoding has been applied to categorical features. It is the process of assigning integer representations (like ‘0’, ‘1’, etc.) to all distinct values within a categorical variable [19].
- **Addition of new feature:** A feature called ‘queue_length’ has been introduced, representing the number of clients waiting for service when a new client enters the queue, as depicted in Figure 2. The creation of new features or transformation of existing ones allows models to be provided with more pertinent information, enabling them to more effectively capture underlying patterns within the data [20].

C. Feature Selection

The dataset used in the study contains many features, but not all are crucial for the prediction of waiting time for a client. Removing unnecessary and redundant variables can reduce the amount of features. Here, we employ stepwise feature selection methods, utilizing both forward selection and backward elimination, to refine the model by systematically adding or subtracting features. The stepwise regression approach enables the identification of the most relevant features while avoiding the computational burden of an exhaustive search across all possible feature combinations [21]. In forward selection, the variables are added in succession. It begins by selecting the variable with the highest correlation with the response scores.

At each step, the variable that produces the largest correlation is attached to the model. Until all variables are incorporated or a threshold is met, the procedure continues to run [22]. On the contrary, backward selection subsequently eliminates variables one at a time whereas all the predictor variables are previously included. At each step, the variable that gives the lowest score is removed from the model. Until only one variable is left or a predetermined ending criterion is satisfied, this procedure keeps going [22].

D. Dimensionality Reduction

The mitigation of the undesirable features of high-dimensional spaces is the reason that dimensionality reduction is useful in numerous fields like data compression, noise reduction, etc [23]–[25]. In the present case, we have employed PCA which is a popular dimensionality reduction technique that can remove extraneous noise and data while retaining a significant amount of information [26]. Following the acquisition by PCA analysis, a stepwise feature selection procedure (discussed in subsection C) has been employed to figure out the optimal set. Such tuning involves the application of both forward selection and backward elimination methods.

E. Model Selection

Previously, a few studies have approached waiting time prediction as a classification task, as their prediction models yielded discrete outputs [2], [27]. However, our study aims to predict waiting time as a continuous numerical output, treating it as a regression task. Since this is a regression problem, ensemble methods are chosen for this task. Ensemble learning initially extracts a set of features through diverse transformations. Using these features, multiple learning algorithms are applied to generate preliminary predictions. The ensemble then combines the information from these predictions, employing adaptive voting schemes to facilitate knowledge discovery and improve overall predictive performance [28], [29]. Both theoretically and empirically, ensemble methods have been demonstrated to outperform single predictors across a wide range of tasks [30]. To start with, RF combines the concepts of bagging and random subspace [31]. It's a powerful predictor due to its ability to avoid overfitting. By introducing appropriate randomness, it excels at classification and regression tasks [32]. Extra Trees was introduced as a computationally efficient and highly randomized extension of the random forest algorithm [33], [34]. Lastly, the GBR minimizes a specific loss function with weight distribution adjustments, resulting in the sequential growth of trees. By using forward stage-wise modeling and averaging, it lessens variance and model bias, respectively [35]. HGBR is another histogram-based gradient-boosting method that delivers quicker training and less memory consumption [36]. As part of an ensemble approach, the VR creates many models and aggregates their forecasts to provide more accurate predictions [31]. In contrast to bagging, where base models are trained on different subsets of the data, VR involves training each base model on the entire dataset. The final prediction is derived from the combined outputs of all the models [37].

F. Model Training

In this study, the training process involves a systematic approach through splitting the dataset, ensuring reproducibility, and leveraging the capabilities of the scikit-learn library [38]. The dataset is divided into training and test sets using an 80:20 split ratio. This indicates that 80% of the data is allocated for the model training and 20% is left reserved for the performance evaluation. To maintain consistency and reproducibility, a random state of 42 has been employed ensuring that the data-splitting process produces the same results each time when the code is executed.

III. EVALUATION METRICS

Evaluation metrics are numerical measurements that evaluate an algorithm, model, or system's quality and performance. They are essential for figuring out how a model performs on a particular task and for drawing accurate conclusions from performance.

A. R^2 Score (Coefficient of Determination):

Based on the independent variables, the variance of the dependent variable can be predicted, and the coefficient of determination can be represented as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (X_i - Y_i)^2}{\sum_{i=1}^n (\bar{Y} - Y_i)^2} \quad (1)$$

Here, X_i indicates the probable i th value, and Y_i denotes the true i th value, where $\bar{Y}$ represents the mean of Y_i values.

$$\bar{Y} = \frac{1}{m} \sum_{i=1}^m Y_i$$

The coefficient of determination assumes values between the worst possible value $-\infty$ to the best possible value $+1$, depending on the mutual relationship between the ground truth and the prediction model. R-squared offers an indication of the goodness of fit, assessing the extent to which the observed values align with the predicted values in the regression model [39].

B. Mean Squared Error (MSE)

The squared difference between the actual and expected values is quantified by MSE. Essentially, it measures the proximity of the line of best fit to the dataset. As a positive value, it is obtained by squaring the differences to eliminate negative signs. A smaller MSE indicates a more accurate prediction, as it signifies a closer alignment between the predicted and actual values [40].

$$MSE = \frac{1}{m} \sum_{i=1}^m (X_i - Y_i)^2 \quad (2)$$

C. Root Mean Squared Error (RMSE)

RMSE is derived as the square root of the mean of the squares of all the errors. The standard deviation corresponds to what RMSE stands for. Similar to MSE, RMSE assesses the proximity of the line of best fit to the dataset, providing insight into the accuracy of the prediction [40].

$$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (X_i - Y_i)^2} \quad (3)$$

D. Mean Absolute Error (MAE)

The mean of the absolute differences between the predicted and actual values is designated as MAE. Unlike MSE, which computes the squared differences, MAE measures the absolute distinctions [40].

$$MAE = \frac{1}{m} \sum_{i=1}^m |X_i - Y_i| \quad (4)$$

E. Accuracy

The accuracy metric calculates the proportion of accurate predictions to the number of examined instances [41].

$$\text{Accuracy} = \frac{tp + tn}{tp + fp + tn + fn} \quad (5)$$

Here, tp and tn signify the number of accurately classified positive and negative instances respectively. Concurrently, fp and fn indicate the count of misclassified positive and negative events, respectively.

F. F1 Score

The statistic that strikes a compromise between precision and recall is the F1 Score, where precision detects the share of accurately identified objects among the predicted objects and recall signifies the percentage among the actual objects. The F1 Score is beneficial in analyzing imbalanced datasets where one class may remain more frequent than the others. It ranges from 0 to 1, representing poor performance to perfect precision and recall, respectively [42].

$$F1 = \frac{2tp}{2tp + fp + fn} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (6)$$

IV. RESULT AND DISCUSSION

This section evaluates the overall performance of different models based on the performance metrics along with a comparison of various data preprocessing techniques. Additionally, the outcomes of the classification approach have been assessed.

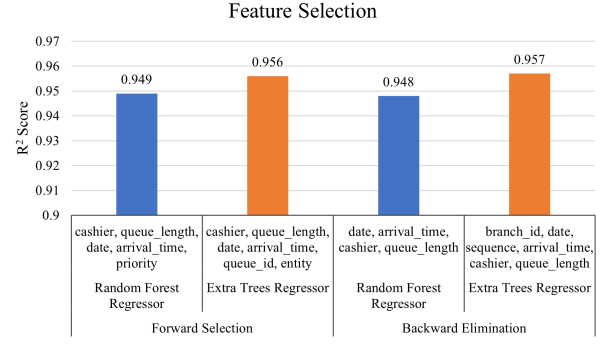


Fig. 3: Top Scoring Feature Set Based on R² score

A. Performance

Here, we evaluate the effectiveness of feature selection methods, dimensionality reduction, and multiple regression models to establish the most accurate approach for predicting waiting times. This analysis underscores the importance of key features and contrasts the behavior of different machine learning models.

Figure 3, presents the top-scoring feature sets identified by two feature selection methods, using two different models. In both forward selection and backward elimination, the ET regressor slightly outperformed the RF regressor in terms of R² score. The consistency of four features: ‘cashier’, ‘queue_length’, ‘date’, and ‘arrival_time’ in both forward and backward selection methods underscore their significance. This finding is practical, as the service time is greatly influenced by the efficiency of the service provider, in this case, the ‘cashier’, which significantly impacts the waiting time [43]. In addition, the ‘queue_length’ directly affects the waiting time, as longer queues naturally lead to increased waiting periods [44]. The ‘date’ is crucial because the demand for services can vary significantly on different days [45], impacting the overall waiting time. Similarly, ‘arrival_time’ is also important because certain hours of the day may experience higher client volume [46], influencing the waiting period.

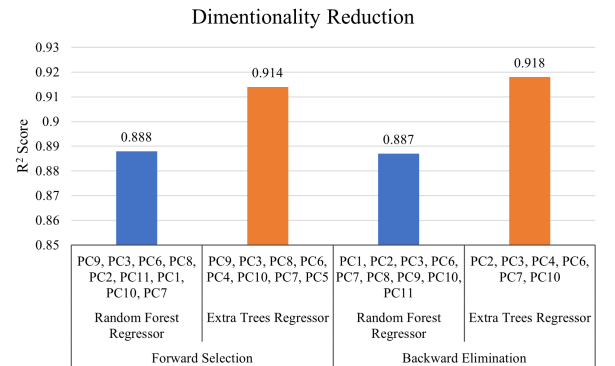


Fig. 4: Top-scoring Principal Component sets based on R² score

In Figure 4, the ET regressor again performed better than the RF regressor in the dimensionality reduction using PCA. However, the widely-used dimensionality reduction technique,

PCA fails to outperform the original feature sets with unsatisfactory R^2 scores, indicating that dimensionality reduction is not effective in this case. Notably, feature selection methods such as forward selection and backward elimination are instrumental in identifying the most effective feature sets. The feature set that yields the best R^2 score in these methods has been chosen for waiting time prediction. Hence, further performance evaluation has been conducted using the most significant identified features in conjunction with several ensemble tree-based regression algorithms.

Models	R^2 Score	MAE	MSE	RMSE
Random Forest Regressor	0.949	3.67	34.44	5.86
Extra Trees Regressor	0.957	3.06	29.82	5.46
Gradient Boosting Regressor	0.800	7.28	140.29	11.84
Hist Gradient Boosting Regressor	0.900	5.28	62.35	7.89
Voting Regressor	0.860	6.07	95.32	9.76

TABLE II: Model-wise performance metrics

The performance metrics for several models are presented in Table 2. The ET regressor emerges as the top performer, showcasing the lowest MAE, MSE, and RMSE scores of 3.06, 29.82 and 5.46 minutes along with the highest R^2 score. Additionally, both the RF regressor and ET regressor exhibit closely comparable performance across all metrics examined in the study. Conversely, the GBR ranks as the lowest performer among the five models, displaying inferior results across all metrics.

B. Classification Strategy

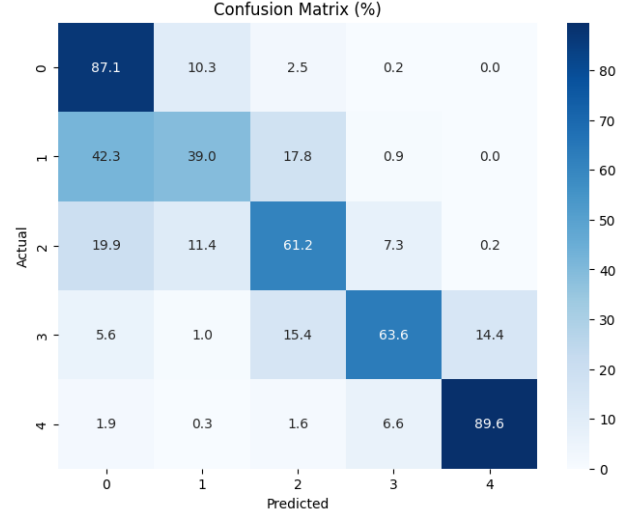
For the classification approach, we categorize waiting times into five classes: very short (0-5 mins), short (6-10 mins), normal (11-20 mins), long (21-30 mins), and very long (>30 mins). But the dataset has an issue with class imbalance as shown in Table 3.

Class	Class Label	Class Counts
Very Short (0-5 mins)	0	1593
Short (6-10 mins)	1	5792
Normal (11-20 mins)	2	2212
Long (21-30 mins)	3	2092
Very Long (>30 mins)	4	1019

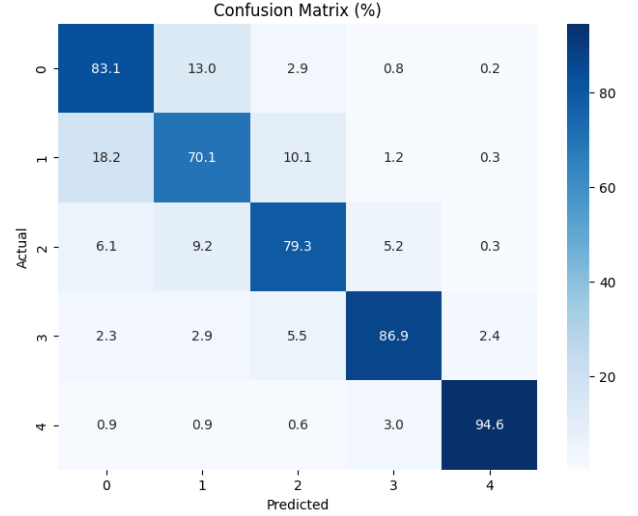
TABLE III: Class-wise distribution of waiting times

The class is imbalanced if not approximately equally represented in the dataset. Our approach to such a problem by under-sampling the classes larger than the minority class reduces the dataset's size, negatively impacting performance. On the contrary, an alternative method called Synthetic Minority Over-sampling Technique (SMOTE) unlike traditional over-sampling with replacement, generates synthetic examples to over-sample the minority class, thereby enhancing the balance without reducing the overall dataset size [47].

In the classification technique, the ET classifier outperforms other models. Figure 5(a), presents the confusion matrix for the ET classifier's results in predicting waiting times using the original dataset. This matrix indicates significantly poor prediction accuracy in classes 1, 2, and 3. Balancing the data in



(a) Confusion matrix of the original dataset



(b) Confusion matrix after using SMOTE

Fig. 5: Confusion Matrices

Methods	Models	Time overflow		Regression	
		Accuracy	F1 Score	R2 Score	MAE
Carvalho et al.	GBM	0.97	0.75	-	-
	RF	0.96	0.68	-	-
Ours	GBM	0.97	0.97	0.949	3.67
	RF	0.97	0.97	0.8	7.28
	ET	0.97	0.97	0.957	3.06

TABLE IV: Performance comparison

this manner leads to a significant improvement in the classifier model's performance, enhancing its ability to predict waiting times accurately, as displayed in Figure 5(b).

C. Performance Comparison

In this section, we present a comparative performance analysis between our study and that of Carvalho et al., utilized the same dataset. We examine both time overflow prediction and regression analysis, highlighting key differences and improvements with our approach.

From Table 4, it is evident that in time overflow prediction, Carvalho et al. achieved their best result using the GBM model, with an accuracy of 0.97 and an F1 score of 0.75 [2]. In contrast, our study attains the same best accuracy of 0.97 using both RF and GBM models, and an improved F1 score of 0.97 for both models, which is an indication of superior performance in handling imbalanced data and predicting precise waiting times. While the previous study did not explore any regression analysis, the current approach obtains optimal results using the ET regressor model, attaining an R^2 score of 0.957 and an MAE of 3.06 minutes.

V. CONCLUSION

Unlike previous approaches, this work gives a complete approach concentrating on the exact assessment of the target variable with a target to accurately anticipate waiting times. Ensemble tree algorithms have been selected for their superior performance. Through meticulous feature selection, the highest R^2 scores have been found in identified key features: 'cashier', 'queue_length', 'date', and 'arrival_time'. Ensemble tree models outperform traditional methods, underscoring their ability to capture non-linear relationships in the data. Taking PCA into account, the original feature sets become more effective. In addition to regression analysis, a classification approach by segmenting waiting times into distinct groups has also been explored. To address class imbalance, the utilization of SMOTE significantly improves classification accuracy, particularly for minority classes. The overall improvement can be attributed to the effective integration of SMOTE and the robust performance of the ET regressor. Although this study faces challenges due to limited and inadequately documented datasets, it effectively demonstrates the efficacy of ensemble methods in predicting waiting times. Future research should prioritize the acquisition of more comprehensive datasets and the exploration of advanced feature engineering.

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